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Clamping Device for Fastening a Component to a Bicycle Handlebar

Abstract

A clamping device for fastening a component to a bicycle handlebar includes (i) an open, annular base body bent around a central axis having a first circumferential end and a second circumferential end, and (ii) a locking area configured to form a detachable snap connection with the component. The base body is configured to enclose the bicycle handlebar. A joining gap is formed between the first circumferential end and the second circumferential end. The locking area is arranged at the first circumferential end. The first circumferential end and the second circumferential end are configured to be connected by a fastening element along a fastening axis in order to fix the clamping device to the bicycle handlebar. The locking area is configured to fix the component by way of the fastening element.

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Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 201 907.0, filed on Mar. 1, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.

Background

[0002] The present disclosure relates to a clamping device for fastening a component to a bicycle handlebar, a clamping assembly and a method for fastening a component to a bicycle handlebar.

[0003] Clamping devices for fastening components to a bicycle handlebar are known in the prior art in various embodiments. In order to fasten a component to a bicycle handlebar with the aid of a clamping device according to the prior art, a complex assembly process is usually necessary. This may comprise a variety of assembly steps. Often, an extensive alignment of the components with respect to one another is also necessary.

[0004] It would be desirable to have a clamping device for fastening components to a bicycle handlebar, which allows for intuitive and fast assembly of components on a bicycle handlebar while being simple and inexpensive to manufacture.

SUMMARY

[0005] The clamping device according to the disclosure for fastening a component to a bicycle handlebar with the features set forth below has the advantage that the component can be easily aligned and fastened in a locking area of the clamping device, wherein the clamping device and the component can then be fixed to the bicycle handlebar with a fastening element. According to the present disclosure, this is achieved by the clamping device comprising an open, annular base body bent about a central axis with a first circumferential end and a second circumferential end, which is configured to enclose the bicycle handlebar. A joining gap is configured between the first circumferential end and the second circumferential end. Furthermore, the clamping device comprises a locking area, which is arranged at the first circumferential end and is configured to form a detachable connection with the component. The detachable connection is in particular a detachable snap connection. The first circumferential end and the second circumferential end are configured to be connected by at least one fastening element, preferably precisely one screw, along a fastening axis in order to fix the clamping device to the bicycle handlebar. Furthermore, the locking area is configured to fix the component by way of the at least one fastening element in addition to the detachable connection. The fastening axis is preferably arranged perpendicular to the central axis. The component can thus be temporarily aligned and fastened to the clamping device with the aid of the snap connection. The clamping device and the component can then be quickly and easily fastened to the bicycle handlebar permanently with the fastening element. In particular, the clamping connection according to the disclosure enables the one-handed fixation of the component to the bicycle handlebar.

[0006] Preferred refinements of the disclosure are set forth below.

[0007] Preferably, the locking area has an undercut and a protrusion, which are configured to form the detachable snap connection with the component. The undercut and the protrusion are preferably configured in such a way that, in combination, they restrict all degrees of freedom of the component to the clamping device in order to form a positive-locking connection with the component. The protrusion is preferably resilient and forms a snap element to allow for ease of joining and detaching the snap connection. The combination of undercut and protrusion allows for a snap connection, which can be easily integrated into the clamping device and which can be used to easily fasten components to the clamping device.

[0008] Further preferably, the protrusion from the base body is arranged radially outside the fastening axis at the first circumferential end and the undercut is arranged between the central axis of the base body and the fastening axis. This allows for easy fastening and subsequent fixation of the component to the clamping device.
[0009] The first circumferential end and the second circumferential end preferably have through-holes to receive the screw. Through-holes may be manufactured inexpensively. The screw preferably engages in a thread, which is attached to the component in order to fasten the component and the clamping device to the bicycle handlebar. The through-hole in the first circumferential end is preferably of greater diameter than the through-hole in the second circumferential end.
[0010] According to a further preferred embodiment of the disclosure, the first circumferential end and/or the second circumferential end have a spacer, which is arranged radially outside the fastening axis from the base body and protrudes into the joining gap. Thus, the spacer first closes the contact between the first circumferential end and the second circumferential end, wherein the joining gap with a defined distance is still present between the first circumferential end and the second circumferential end. The joining gap may be reduced during the fastening operation with the aid of the screw, thereby ensuring reliable application of a clamping force.
[0011] The clamping device preferably comprises a guide bar, which is arranged adjacent the undercut and forms a lateral stop in the direction along the central axis. The clamping device can thus enable the simple attachment and simultaneous alignment of the component.
[0012] Further preferably, the clamping device is one-piece. Thus, the clamping device can be manufactured inexpensively in one step. Furthermore, the one-piece clamping device allows for reliable use of the clamping device.
[0013] Further, the disclosure describes a clamping assembly comprising a fastening element, a component, and a clamping device previously described. The clamping device is configured to fix the component to a bicycle handlebar by way of the fastening element. The fastening element is preferably precisely one screw. This enables a simple and reliable fastening of the component with the aid of the clamping device to the bicycle handlebar.
[0014] Further preferably, the clamping assembly is configured to fix the component below the bicycle handlebar. Thus, the center of gravity of the clamping assembly is positioned below the bicycle handlebar, allowing a stable position to fix the clamping assembly.
[0015] Furthermore, the disclosure comprises a method for fastening a component to a bicycle handlebar. In one step, a clamping device previously described is attached to a bicycle handlebar. In a further step, the component is fastened to the clamping device. In a final step, the clamping device is fixed to a bicycle handlebar and the component is fixed to the clamping device with a fastening element. The fastening element is in particular precisely one screw. Thus, the method allows a simple and reliable fastening of the component to the bicycle handlebar. The component may be fastened to the clamping device before or after the clamping device is attached to the bicycle handlebar.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0016] A preferred exemplary embodiment of the disclosure is explained in detail below with reference to the accompanying drawings. The drawing shows:
[0017] FIG. **1** a schematic perspective view of a clamping device according to a first preferred exemplary embodiment of the disclosure,
[0018] FIG. **2** a detailed view of the first circumferential end of the clamping device according to the first preferred exemplary embodiment of the disclosure,
[0019] FIG. **3** a sectional view of the clamping device according to the first exemplary embodiment

and a bicycle handlebar before the clamping device is attached to the bicycle handlebar,
[0020] FIG. **4** a sectional view of the bicycle handlebar and the clamping device according to the first exemplary embodiment after the clamping device has been attached to the bicycle handlebar,
[0021] FIG. **5** a sectional view of a clamping assembly according to the first preferred
[0022] exemplary embodiment of the disclosure from the clamping device and a component fixed to the bicycle handlebar with the aid of a screw,
[0023] FIG. **6** a detailed view of the clamping assembly according to the first preferred exemplary embodiment of the disclosure in the area of the joining gap, and
[0024] FIG. **7** a flow diagram of a method for fastening the component to the bicycle handlebar according to the first exemplary embodiment of the disclosure.
DETAILED DESCRIPTION
[0025] Preferably, all identical components, elements, and/or units are provided with the same reference symbols in all figures.
[0026] Referring to FIGS. **1** to **7**, a clamping device **10**, a clamping assembly **100**, and a method for fastening a component **40** to a bicycle handlebar **50** according to a first exemplary embodiment will be described in detail below.
[0027] FIG. **1** shows a perspective view of the clamping device **10**. The clamping device **10** comprises an open annular base body **12** bent about a central axis X-X. The base body **12** has a first circumferential end **14** and a second circumferential end **16**. The clamping device **10** has a joining gap **18** between the first circumferential end **14** and the second circumferential end **16**.
[0028] The first circumferential end **14** and the second circumferential end **16** each have a flat joining surface **27**. The joining surfaces **27** are aligned parallel to the central axis X-X.
[0029] The first circumferential end **14** and the second circumferential end **16** each have a through-hole **26** that is aligned perpendicular to the central axis X-X. The through-holes **26** are arranged within and perpendicular to the joining surface **27**.
[0030] In the fastened state, the through-holes **26** have a common fastening axis S-S. In FIG. **1**, the fastening axis S-S is shown in the through-hole **26** at the first circumferential end **14**.
[0031] The second circumferential end **16** has a spacer **28** that protrudes from the second circumferential end **16** towards the joining gap **18**. The spacer **28** is arranged radially outside the through-bores **26** when viewed from the central axis X-X.
[0032] The clamping device **10** is preferably an injection-molded component made of a plastic material. Thus, this can be manufactured quickly and inexpensively. Furthermore, plastic has good resiliency, which allows for simple fastening of the clamping device **10** to the bicycle handlebar **50** and the component **40** to the clamping device **10**.
[0033] The first circumferential end **14** has an outwardly directed locking area **20**. The locking area **20** is configured to form a detachable snap connection with the component **40**. For this purpose, the locking area **20** has an undercut **22** and two cuboid protrusions **24**, which can form the snap connection on the component **40** with a corresponding indentation **41** (shown in FIGS. **5** and **6**) and lug **43** (shown in FIG. **5**). The lug on the component **40** can engage with the undercut **22**. By subsequently rotating the component **40**, the protrusions **24** can snap into the indentations **41** of the component **40**, thereby fastening the component to the clamping device.
[0034] In the locking area **20**, a guide bar **23** is arranged next to the undercut **22** on both sides, such that the undercut **22** is centrally located between the guide bars **23**. The guide bar **23** extends along an orthogonal plane to the central axis X-X. Thus, the guide bar **23** forms a guide and a stop for the component **40**.
[0035] FIG. **2** shows a detailed view of the locking area **20**.
[0036] The locking area **20** has a flat stop surface **29**, which is arranged parallel to the joining surface **27**. The through-hole **26** extends from and is arranged perpendicular to the stop surface **29** to the joining surface **27**.
[0037] At a radial end of the locking area **20**, the two cuboid protrusions **24** are arranged on the

stop surface **29**. The protrusions **24** have a chamfer **21**, which facilitates the formation and detachment of a snap connection with the component **40**. A screw **60** may be used to fix the component **40** to the stop surface **29** and to the protrusions **24**.

[0038] FIG. **3** shows a sectional view of the clamping device **10** and the bicycle handlebar **50**. The clamping device **10** is not yet attached to the bicycle handlebar **50**. The bicycle handlebar **50** has a circular cross-section.

[0039] To attach the clamping device **10** to the bicycle handlebar **50**, the first circumferential end **14** and the second circumferential end **16** are bent apart such that the joining gap **18** increases. The base body **12** has such resilience that the first circumferential end **14** and the second circumferential end **16** can be bent apart so far that the joining gap **18** is wider than the diameter of the bicycle handlebar **50** without the base body **12** deforming plastically.

[0040] The second circumferential end **16** has a recess **25** on a side opposite the joining gap **18**, which is configured to receive a screw head of the screw **60**. The recess **25** is configured such that the screw head can rest flush against it.

[0041] FIG. **3** shows a basic state of the clamping device **10**, in which the joining gap **18** is not bent open or closed by any external forces.

[0042] In FIG. **4**, the clamping device **10** can be seen after being attached in the bicycle handlebar **50**. In doing so, the clamping device **10** has returned to its basic state.

[0043] In the attached basic state, there is a clearance mount between the base body **12** and the bicycle handle **50**, such that the clamping device **10** is freely movable along the bicycle handlebar **50**.

[0044] The locking area **20** is geodetically arranged below the central axis X-X of the clamping device **10**.

[0045] FIG. **5** shows the clamping assembly **100** consisting of screw **60**, component **40** and clamping device **10**. The clamping assembly **100** is fixed to the bicycle handlebar **50** with the aid of the screw **60**.

[0046] For example, the component **40** is shown as a bicycle light, but may also be, for example, a bicycle computer, a bell, an additional sensor or an input unit.

[0047] The component **40** forms a detachable snap connection with the locking area **20**. The lug **43** of the component **40** thereby engages in the undercut **22** of the clamping device **10**. Furthermore, the protrusion **24** of the clamping device **10** engages in the associated indentation **41** of the component **40**.

[0048] The component **40** is fixed to the clamping device **10** with the aid of the screw **60**. The screw extends through the through-holes **26** in the first circumferential end **14** and the second circumferential end **16** and engages in an associated threaded hole **42** in the component **40**. The threaded hole **42** is arranged coaxially to the through-holes **26**. The screw head abuts the recess **25** of the second circumferential end **16** such that the screw **60** contracts the second circumferential end **16** and the component **40**.

[0049] The second circumferential end **16** abuts the spacer **28** against the joining surface **27** of the first circumferential end **14**. There is also a narrow joining gap **18** between the joining surface **27** of the first circumferential end **14** and the joining surface **27** of the second circumferential end **16**. By screwing the screw **60** further into the component **40**, the joining gap **18** can be reduced, which increases the clamping force of the clamping device **10** on the bicycle handlebar **50**.

[0050] FIG. **6** shows a detailed view of the clamping assembly **100** of FIG. **5** in the area of the joining surface.

[0051] It can be seen that the through-hole **26** in the first circumferential end **14** has a larger diameter than the through-hole **26** in the second circumferential end **16**. Furthermore, the through-hole **26** in the first circumferential end **14** has a chamfer **21**, which is adjacent the joining gap **18**. The larger diameter and chamfer **21** in the first circumferential end **14** facilitate insertion of the screw **60** when the through-holes **26** in the first and second circumferential ends **14**, **16** are non-

coaxially aligned with each other.
[0052] FIG. **7** shows a flow chart of a method for fastening the component **40** to the bicycle handlebar **50**.
[0053] In a first step S**1**, the clamping device **10** is attached to the bicycle handlebar **50**. To this end, the clamping device **10** is preferably bent open to such an extent that the joining gap **18** is wider than the diameter of the bicycle handlebar **50**. After the clamping device **10** has been attached to the bicycle handlebar, it preferably bends back to the base state.
[0054] In a second step S**2**, the component **40** is fastened to the clamping device **10** by way of a detachable snap connection.
[0055] Alternatively, the component **40** may also be fastened to the clamping device **10** before the clamping device **10** is fastened to the bicycle handlebar **50**.
[0056] Finally, in a third step S**3**, the clamping device **10** is fixed to the bicycle handlebar **50** and at the same time the component **40** is fixed to the clamping device **10** with precisely one screw **60**. Thus, with the aid of the clamping device **10**, the component **40** can be fixed to the bicycle handlebar **50** in a few steps.

Claims

1. A clamping device for fastening a component to a bicycle handlebar, comprising: an open annular base body bent about a central axis and having a first circumferential end and a second circumferential end; a locking area configured to form a detachable connection with the component; and wherein the base body is configured to enclose the bicycle handlebar, wherein, in the assembled state, a joining gap is defined between the first circumferential end and the second circumferential end, wherein the locking area is arranged at the first circumferential end, wherein the first circumferential end and the second circumferential end are configured to be connected by at least one fastening element along a fastening axis in order to fix the clamping device to the bicycle handlebar, and wherein the locking area is configured to fix the component in addition to the detachable connection by way of the at least one fastening element.
2. The clamping device according to claim **11**, wherein the locking area has an undercut and a protrusion which are configured to form the detachable snap connection with the component.
3. The clamping device according to claim 2, wherein: the protrusion from the base body is arranged radially outside the fastening axis at the first circumferential end, and the undercut is arranged between the central axis and the fastening axis.
4. The clamping device according to claim 1, wherein the first circumferential end and the second circumferential end have through-holes which are configured to receive the at least one fastening element.
5. The clamping device according to claim 1, wherein the first circumferential end and/or the second circumferential end have a spacer arranged radially outside the fastening axis from the base body and protruding into the joining gap.
6. The clamping device according to claim 1, further comprising a guide bar arranged adjacent the undercut and forming a lateral stop in the direction along the central axis.
7. The clamping device according to claim 1, wherein the clamping device is formed so as to be one piece.
8. A clamping assembly, comprising: a fastening element; a component; and a clamping device according to claim 1, wherein the clamping device is configured to fix the component to a bicycle handlebar by way of the fastening element.
9. The clamping assembly according to claim 8, wherein the clamping assembly is configured to geodetically fix the component below the bicycle handlebar.
10. A method of fastening a component to a bicycle handlebar, comprising: attaching a clamping device according to claim 1 to the bicycle handlebar; fastening the component to the clamping

device; and fixing the clamping device to the bicycle handlebar and the component to the clamping device with a fastening element.

11. The clamping device according to claim 1, wherein the detachable connection is a detachable snap connection.

12. The clamping device according to claim 1, wherein the at least one fastening element includes precisely one screw.

13. The clamping assembly according to claim 8, wherein the fastening element includes precisely one screw.

14. The method of claim 10, wherein the fastening clement includes precisely one screw.